

Climate-Aware Portfolio Construction: Minimum Variance, Carbon Constraints and Net-Zero Alignment

Ishaq Chekara, Saad Chraibi, Khalil Mokni ,Yazid Abaroudi HEC Lausanne

May 28, 2026

Abstract

This report investigates whether climate-aware portfolio strategies can reduce the carbon footprint of an equity portfolio without reducing financial attractiveness. Using the monthly dataset of Pacific-region stocks from 2013 to 2024 and carbon emissions data, we implemented five portfolios: a value-weighted benchmark (*VW*), a minimum-variance portfolio (*MV*), a minimum-variance portfolio with a 50% carbon cap (*MV_50*), a value-weighted portfolio with, again, a 50% carbon cap (*VW_50*), and another one following a Net-Zero strategy with an annually declining carbon footprint (*VW_NZ*). Portfolios are rebalanced annually and evaluated monthly. We compare financial performance through different metrics, carbon footprint and weighted average carbon intensity (WACI). Our results show that the minimum-variance portfolio outperforms the value-weighted one and that carbon-constrained portfolios achieve large reductions in carbon footprint, although WACI does not always decline, while maintaining competitive returns. However, the constrained portfolios tend to be less diversified, therefore exhibiting higher idiosyncratic risk and tracking error. The Net-Zero strategy tracks the declining carbon cap closely and shows competitive financial results, suggesting that climate-aware investing can be applied in a structured portfolio strategy. The findings highlight important trade-offs between carbon reduction, diversification and benchmark deviation, suggesting that investors need to handle these factors simultaneously.

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1 Introduction

Sustainable investing has become a central theme in finance. Growing awareness of climate-related issues led investors to pay closer attention to companies' exposure to transition risks, including carbon taxes and regulatory penalties). Investors are aware that these risks may be reflected in asset prices. Conventional portfolio construction, which focuses on expected returns and covariances, does not account for climate externalities and may concentrate investments in carbon-intensive firms. The central question of this project is whether investors can reduce the carbon footprint of a portfolio without changing the set of stocks available for investment.

We study this question by constructing five long-only portfolios using monthly stock returns and carbon data for the Pacific region. All of them are rebalanced each year in proportion to market capitalization (except for the benchmark which is rebalanced each month). The portfolios are:

- **Value-weighted benchmark (VW):** a value-weighted portfolio constructed without constraint. This portfolio serves as a benchmark for the project.
- **Minimum-variance portfolio (MV):** a portfolio that minimises total variance.
- **Carbon-constrained minimum-variance portfolio (MV_{50}):** a minimum-variance portfolio that limits the portfolio carbon footprint to 50% of the unconstrained MV carbon footprint.
- **Carbon-constrained benchmark-relative portfolio (VW_{50}):** a portfolio that reproduces VW benchmark while imposing a 50% carbon footprint cap.
- **Net-Zero benchmark-relative portfolio (VW_{NZ}):** a portfolio that follows a declining carbon cap (10% per annum) starting from the benchmark's 2013 carbon footprint.

Throughout the report we compare the financial performance and carbon emissions of these portfolios, discuss their construction and constraints, and interpret the results. A recurring theme is that reducing carbon exposure is not necessarily costly in terms of returns but may increase concentration and idiosyncratic risk.

2 Data and Implementation

2.1 Data sources and universe construction

Our analysis uses monthly data from January 2013 to December 2025 for equities in the Pacific region. For each company we collect market values, monthly prices, and carbon metrics (scope 1 emissions normalised by business value or revenue). A month-end risk-free rate series spanning December 2013 to December 2024 is used to compute excess returns when constructing performance measures.

The investable universe for each allocation year t is defined by selecting all stocks that have complete return and carbon information over the estimation window (occasional missing values are treated in order to keep as many companies as possible). Firms with missing or unreliable price data, very low prices, or stale prices are excluded when they cannot provide meaningful return estimates. Delisted firms are kept in the sample up to the point at which their market data remain available. The ISIN code is used as a unique identifier to match each company across the different CSV files and to merge the corresponding data. The universe is constructed at the end of year t and the portfolio weights optimised using data available up to that date. We implement the resulting weights over the following calendar year $t+1$.

2.2 Data cleaning and preprocessing

The first data-related issue was converting the original XLSX files into CSV files while keeping the data usable for the analysis. The CSV files had to be structured in a wide format, with dates in rows and firm-level data in columns, identified by each company’s ISIN code.

Another difficulty was determining which firms belonged to the assigned market. For example, although China is geographically part of the Pacific region, it is classified as an emerging market and therefore had to be excluded from our assigned market. In addition, the first two letters of some ISIN codes did not always match the country code. For instance, some firms classified under Hong Kong had ISINs starting with “BMG”, which made it necessary to rely at first on the country code information provided in the static file rather than on the ISIN prefix alone. Non-numeric columns such as company names and country code were dropped as we managed to keep only useful companies. Dates are converted to a consistent format, and panels are sorted chronologically. We forward-fill occasional missing prices for short gaps (up to one month) but drop companies with persistent missing observations (less than 30). Returns matrix is computed using simple return formula. Carbon footprints are expressed in tons of CO₂ equivalent per million US dollars invested (tCO₂e/MUSD), and carbon intensity is expressed as tCO₂e per million dollars of revenue. When a firm has only one of these metrics, we retain the available one and record the other as missing. This results in multiple clean datasets used for the portfolio construction.

2.3 Return computation and rebalancing logic

Monthly total returns are computed from adjusted closing prices using the simple returns definition. Portfolio weights are determined at the end of each year t and held constant over the subsequent year $t+1$ (annual rebalancing). The value-weighted benchmark is rebalanced monthly to reflect changes in market capitalisation. By contrast, the benchmark-relative portfolios (*VW_50* and *VW_NZ*) compute active weights $w_t - w_t^{\text{VW}}$ at the annual allocation date and hold these active positions over the following year. Because of this difference in rebalancing frequency, their relative positions gradually drift away from the benchmark, contributing to tracking error.

2.4 Risk estimators

The covariance matrix Σ_t of stock returns is estimated using monthly returns over a rolling window of up to ten years of available history at the allocation date. We apply no shrinkage or factor model, recognising that different estimators could be tested in robustness checks. A small ridge term εI is added to Σ_t for numerical stability (see Section 3.8). For each portfolio we compute ex-ante risk measures such as total variance or tracking error using Σ_t . Ex-post performance metrics (returns, volatility and Sharpe ratio) are computed from realised monthly returns during the implementation period.

3 Methodology

This section presents the optimisation problems used to construct each portfolio, together with the climate metrics and tracking error measures used to assess their performance.

3.1 Value-weighted benchmark

The value-weighted benchmark (VW) assigns weight

$$w_{i,t}^{VW} = \frac{MV_{i,t}}{\sum_j MV_{j,t}}, \quad (1)$$

to stock i at time t , where $MV_{i,t}$ is the market value of stock i at the allocation date. This portfolio serves as the passive benchmark against which we measure active risk. It is rebalanced each month and holds all available stocks.

3.2 Minimum-variance portfolio

The unconstrained minimum-variance portfolio (MV) solves

$$\min_w w' \Sigma_t w \quad \text{subject to} \quad \sum_i w_i = 1, \quad w_i \geq 0 \quad \forall i, \quad (2)$$

where w is the vector of portfolio weights and Σ_t is the estimated covariance matrix at the end of year t . No carbon or tracking-error constraint is added in this formulation.

3.3 Carbon footprint and WACI

The carbon footprint of a portfolio is defined as the weighted sum of corporate emissions per unit of capital invested:

$$CF_p = \sum_i w_i \cdot CF_i, \quad (3)$$

where CF_i denotes the carbon footprint of firm i , measured in tCO₂e/MUSD invested. A related metric is the weighted average carbon intensity (WACI), defined as

$$\text{WACI}_p = \sum_i w_i \cdot CI_i, \quad (4)$$

with CI_i being the carbon intensity of firm i in tCO₂e per million dollars of revenue. WACI is unitless with respect to invested capital and therefore comparable across portfolios. Both metrics are used to evaluate climate exposure in Section 4.

3.4 Carbon-constrained minimum-variance portfolio (MV_50)

To incorporate a climate objective in the MV portfolio, we impose a carbon footprint limit at 50% of the unconstrained MV portfolio's carbon footprint. The optimisation becomes

$$\min_w w' \Sigma_t w \quad \text{subject to} \quad \sum_i w_i = 1, \quad w_i \geq 0, \quad \sum_i w_i CF_i \leq 0.5 \times CF_{\text{MV}}, \quad (5)$$

where CF_{MV} is the carbon footprint of the unconstrained MV portfolio obtained from (2). The carbon limit is binding in most years, resulting in lower carbon exposure but also stronger active tilts and potentially fewer holdings.

3.5 Carbon-constrained benchmark-relative portfolio (VW_50)

The benchmark-relative carbon-constrained portfolio solves

$$\min_w (w - w^{\text{VW}})' \Sigma_t (w - w^{\text{VW}}) \quad (6)$$

subject to

$$\sum_i w_i = 1, \quad w_i \geq 0, \quad \sum_i w_i CF_i \leq 0.5 \times CF_{\text{VW}}, \quad (7)$$

where CF_{VW} is the carbon footprint of the value-weighted benchmark. The objective in (6) minimises the ex-ante tracking error with respect to the benchmark while the carbon constraint in (7) enforces a 50% reduction in footprint. Because the optimisation is benchmark-relative, the resulting portfolio often remains closer to the benchmark than the MV portfolios, albeit with slightly higher total volatility.

3.6 Net-Zero benchmark-relative portfolio (VW_NZ)

The Net-Zero strategy imposes a time-varying carbon cap that declines by a fixed percentage θ each year. Starting from the benchmark's 2013 carbon footprint $CF_{\text{VW},2013}$, the carbon limit in year t is

$$CF_{\text{target},t} = (1 - \theta)^{t-2013} CF_{\text{VW},2013}. \quad (8)$$

In our implementation $\theta = 0.10$, thus a 10% decarbonisation target per year. The optimisation problem is identical to (6) except that the carbon constraint is replaced by $\sum_i w_i CF_i \leq CF_{\text{target},t}$. This creates a progressively tighter carbon cap, and feasibility may become an issue if the universe lacks sufficiently low-emission companies. The realised portfolio carbon footprint and whether it approached the target are analysed in Section (4.4).

3.7 Tracking error

Tracking error (TE) is the standard deviation of the active return $r_p - r_b$, where r_p is the portfolio return and r_b is the benchmark return. Given the covariance matrix Σ_t , the ex-ante monthly tracking error is

$$TE_{\text{monthly}} = \sqrt{(w - w^{\text{VW}})' \Sigma_t (w - w^{\text{VW}})}, \quad (9)$$

and the annualised tracking error is $TE_{\text{annual}} = \sqrt{12} TE_{\text{monthly}}$. Ex-post tracking errors are computed from realised monthly returns over the implementation period. Low tracking error is desirable for our case in order to compare it to the benchmark, but very low TE may limit the room to reduce carbon exposure.

3.8 Implementation details

Several practical choices shape the empirical results and deserve to be made explicit.

Annual out-of-sample loop. Portfolios are rebuilt year by year: for each allocation year Y , the optimiser uses only information available up to the end of Y (returns, market values, emissions, revenues) and the resulting weights are implemented during $Y+1$. This avoids look-ahead bias and matches the natural frequency of the carbon panel, which is annual.

Covariance estimation. The covariance matrix Σ_Y is estimated from the rolling window of monthly returns available at the allocation date (up to ten years of history when available). To stabilise the quadratic program, we add a small ridge term:

$$\Sigma_Y^{\text{ridge}} = \Sigma_Y + \varepsilon I, \quad (10)$$

where ε is a small positive scalar. The ridge does not alter the economic logic of the portfolios but ensures numerical stability when Σ_Y is close to singular, a frequent issue when the number of eligible firms is large relative to the estimation window.

Universe screening: a data-quality filter, not an exclusion rule. The yearly investable universe is built by removing firms with missing carbon data, non-positive revenues, missing market capitalisation, invalid or stale prices, or too few return observations. Importantly, *high-emission firms are not removed mechanically*: they remain eligible and can still receive positive weight if

the resulting portfolio respects the carbon cap. Decarbonisation in our portfolios therefore comes exclusively from the optimisation constraint, not from a best-in-class exclusion screen.

Benchmark construction: *vw_drift* vs *vw*. The annual loop produces *vw_drift*, a value-weighted portfolio formed at the allocation date and left to drift during the implementation year. For the final performance comparison and the computation of tracking error, we use *vw*; the strict, monthly-rebalanced cap-weighted benchmark based on previous-month market capitalisations. The benchmark-relative portfolios (*VW_50* and *VW_NZ*) inherit a structural contributor to ex-post tracking error from this mismatch in rebalancing frequencies.

Risk-free rate alignment. Excess returns and Sharpe ratios are built from monthly portfolio returns and a risk-free rate matched *calendar-month by calendar-month*. Annualised quantities then use the standard $\sqrt{12}$ scaling for volatility and $\times 12$ for arithmetic means. This calendar alignment ensures that excess returns and Sharpe ratios are not biased by mismatched dates after the data-cleaning steps.

4 Empirical Results

In this section we summarise and interpret our results. We first compare the minimum-variance portfolio with the value-weighted benchmark, then analyse the impact of carbon constraints and discuss the Net-Zero strategy.

4.1 Minimum-variance versus value-weighted benchmark

Figure 1 shows the cumulative growth of one Swiss franc invested in the minimum-variance portfolio (*MV*) and the value-weighted benchmark (*VW*) over the period 2014–2025. Both portfolios generate substantial cumulative gains, but the minimum-variance portfolio performs better overall, reaching about 2.73 times the initial capital versus roughly 2.18 for the benchmark. As reported in Table 1, *MV* delivers an annualised return of 9.14%, exceeding the benchmark’s 7.35%. The volatility of *MV* (11.99%) is lower than that of *VW* (12.89%), producing a return-to-volatility ratio of 0.76 versus 0.57 and an excess Sharpe ratio of 0.62 versus 0.43. These results suggest that a minimum-variance strategy can exploit return covariance structure to achieve superior risk-adjusted performance without any explicit carbon considerations.

Nevertheless, the unconstrained *MV* portfolio has a much higher carbon footprint and WACI than the benchmark. The average carbon footprint of *MV* is around 441 tCO₂e/MUSD invested, compared with 173 tCO₂e/MUSD for *VW*, and its WACI is roughly 1100 tCO₂e/MUSD revenue versus 285 for *VW* (see Table 2). Since we do not penalise environmental exposure, the *MV* optimisation may concentrate the portfolio in a few low-volatility but high-emission utilities.

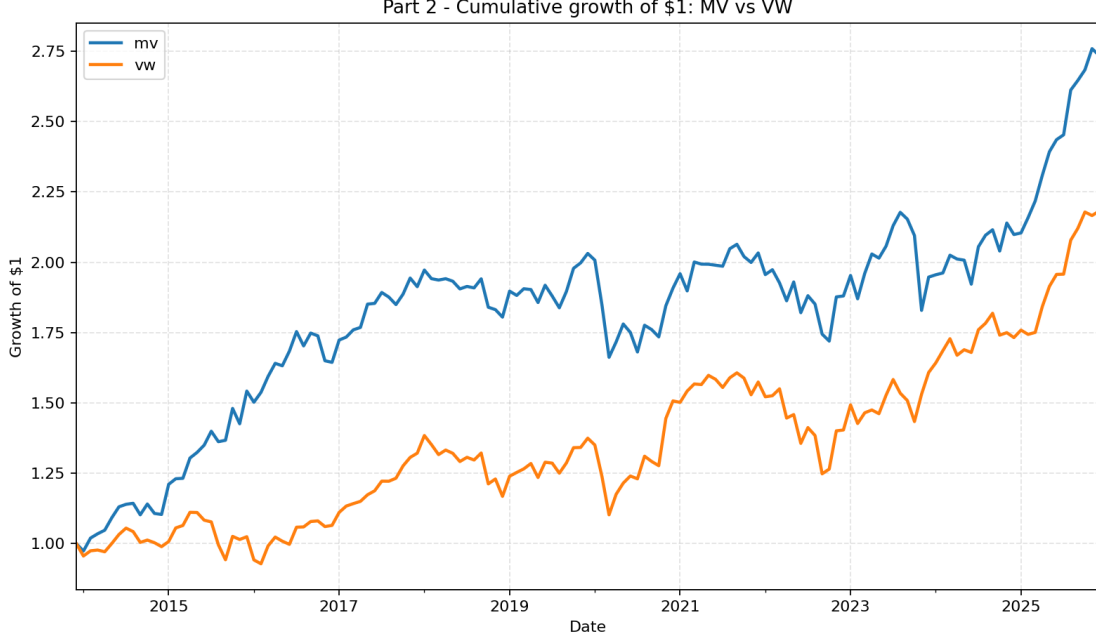


Figure 1: Cumulative growth of one Swiss franc invested in the minimum-variance portfolio (MV) and the value-weighted benchmark (VW) over 2014–2025. The MV portfolio is rebalanced annually, whereas VW is rebalanced monthly.

4.2 Impact of carbon constraints on the minimum-variance portfolio

Introducing a carbon constraint substantially reduces the carbon footprint of the minimum-variance portfolio with only a moderate financial cost. Figure 2 compares the cumulative growth of MV and the 50% carbon-constrained minimum-variance portfolio (MV_{50}). The constrained portfolio not only meets the 50% carbon reduction target but also ends the sample with a performance comparable to the non-constrained one. As shown in Table 1, MV_{50} exhibits slightly lower annualised return (8.88%) than MV , a slightly higher volatility (12.75%) relative to MV , and a return-to-volatility ratio of 0.70. The excess Sharpe ratio of MV_{50} (0.56) exceeds VW .

Table 2 shows that the average carbon footprint of MV_{50} (221 tCO₂e/MUSD) is half that of MV , as intended. The WACI of MV_{50} is slightly higher than that of MV , reflecting the fact that reducing total emissions forces the optimiser to allocate to fewer stocks which may still have high carbon intensities. Noteworthy : the carbon constraint may increase effective concentration and idiosyncratic risk, which should be verified using concentration measures such as the effective number of holdings or the Herfindahl index. Although the metrics seem fairly comparable, both portfolios were constructed using the same universe, meaning that even the investable universe of the MV benchmark was restricted to firms with available carbon data. Furthermore, the carbon emissions of the MV benchmark are highly unstable. Even with the 50% carbon reduction constraint, we cannot conclude that MV_{50} is truly carbon-efficient. (Table 3).

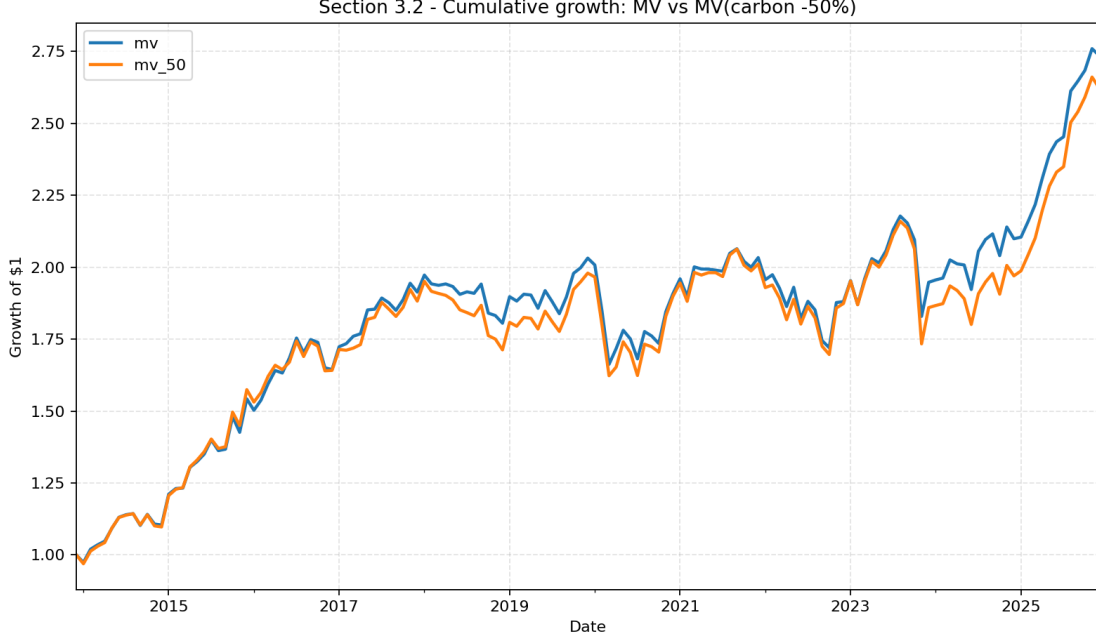


Figure 2: Cumulative growth of the unconstrained minimum-variance portfolio (MV) and the carbon-constrained minimum-variance portfolio (MV_{50}). The carbon-constrained portfolio reduces the carbon footprint by half while maintaining a cumulative performance close to the unconstrained portfolio over 2014–2025.

4.3 Benchmark-relative carbon constraints

Figure 3 displays the cumulative growth of the benchmark and the 50% carbon-reduced portfolio (VW_{50}). The constrained portfolio delivers a similar cumulative return to the benchmark. Table 1 shows that VW_{50} yields an annualised return of 7.36% with volatility of 13.45%. Its return-to-volatility ratio (0.54) and excess Sharpe ratio (0.42) are similar to the benchmark but inferior to MV_{50} .

The average carbon footprint (84tCO₂e/MUSD) is roughly 52% lower than the benchmark's, while WACI rises to 355tCO₂e/MUSD revenue (Table 2). This difference highlights that reducing financed emissions does not necessarily reduce emissions intensity relative to revenues. The portfolio achieves decarbonisation mainly through reallocations across firms rather than through a broad reduction in sector exposure.

Despite the substantial reductions in carbon footprint, the strategy maintains a very low tracking error relative to the benchmark, around 2.9% ex-post. This indicates that benchmark-relative carbon constraints can significantly reduce portfolio emissions while preserving the overall risk-return profile and diversification characteristics of the benchmark.

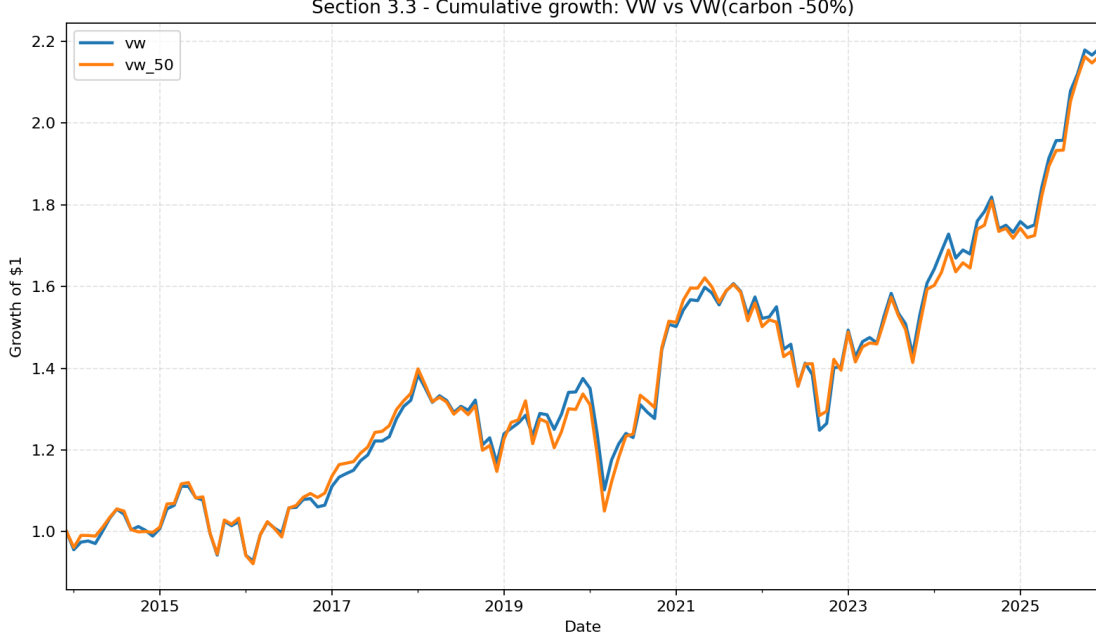


Figure 3: Cumulative growth of the value-weighted benchmark (VW) and the 50% carbon-constrained benchmark-relative portfolio (VW_{50}). The constrained portfolio shows competitive performance despite the carbon cap.

The benchmark-relative constraint tends to produce smaller tracking error than the MV portfolios because the objective explicitly minimises active risk. As reported in Table 3, the average ex-ante tracking error of VW_{50} is around than 0.1%, although ex-post tracking error during implementation rises to about 2.9%. This divergence illustrates that covariance estimates at the allocation date may underestimate future active risk. Moreover, the constrained portfolio may be more concentrated than the benchmark in effective-weight terms, making it more susceptible to idiosyncratic and sector risk. While the carbon reduction appears to come “for free” in terms of realised return, it is nuanced by higher concentration and less diversification.

4.4 Net-Zero benchmark-relative strategy

The Net-Zero portfolio (VW_{NZ}) imposes a carbon cap that shrinks by $\theta = 10\%$ each year, anchored on the benchmark’s 2013 carbon footprint. Figure 4 plots the target cap and the realised carbon footprint of the portfolio. The realised footprint remains below the target throughout the period, showing that the Net-Zero constraint is feasible in each allocation year.

The constraint is binding or nearly binding in several years, especially when the realised footprint lies close to the target cap. In other years, the realised footprint falls clearly below the cap, meaning that the portfolio can achieve a lower carbon footprint than required without sacrificing the benchmark-relative structure. This pattern is consistent with the optimisation logic: when the carbon cap is binding, the optimiser sits at the limit; when other constraints or the available low-carbon investment universe dominate, the realised footprint remains below the target.

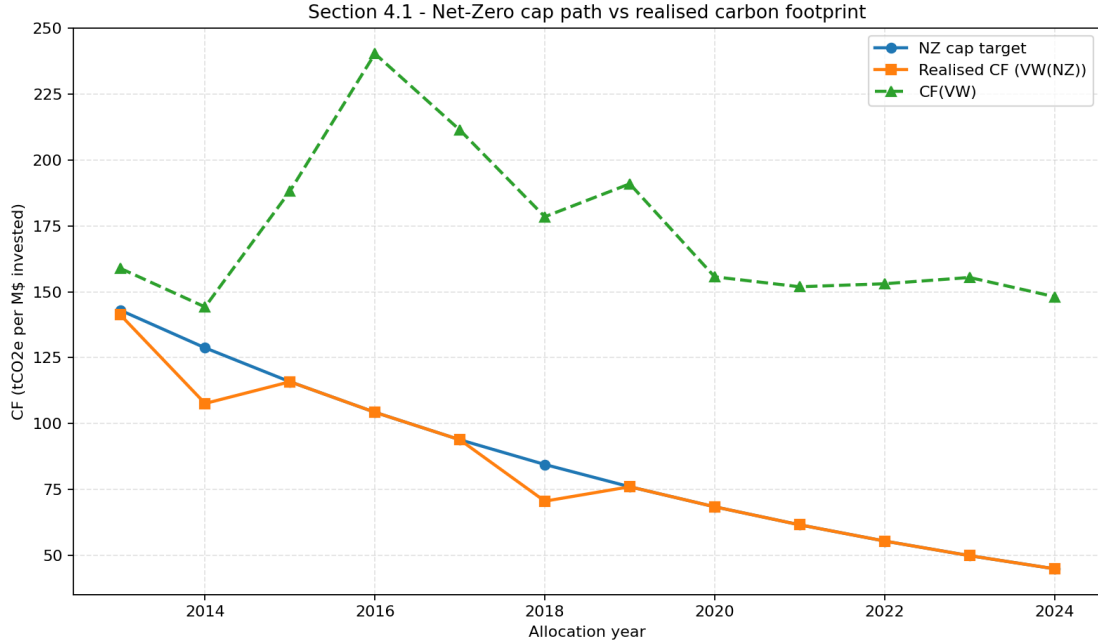


Figure 4: Net-Zero carbon cap (target) and realised carbon footprint of the Net-Zero portfolio (VW_NZ). The target declines by 10% each year from the benchmark’s 2013 carbon footprint, and the realised footprint stays at or below the limit in every year.

Figure 5 compares the cumulative growth of the benchmark portfolio (VW) with the carbon-constrained strategies VW_50 and VW_NZ. The three trajectories remain extremely close over the full 2014–2025 period, indicating that substantial reductions in financed carbon emissions can be achieved without altering the portfolio’s long-run performance profile.

The benchmark-relative climate strategies neither significantly outperform nor underperform the benchmark, but they achieve meaningful carbon reductions with only limited active risk. As reported in Table 1, VW_NZ delivers an annualised return of 7.50% with volatility of 13.49%, corresponding to a return-to-volatility ratio of 0.56 and an excess Sharpe ratio of 0.43. These figures remain very close to those of VW_50, confirming that progressively tightening carbon constraints do not substantially deteriorate risk-adjusted performance.

The ex-post tracking error of VW_NZ is approximately 2.9%, almost identical to VW_50’s 2.8%. This relatively low tracking error suggests that the decarbonisation objective is implemented mainly through moderate reallocations around benchmark weights rather than through large deviations in sector or factor exposures. Overall, the results support the idea that benchmark-relative climate investing can deliver significant emissions reductions while keeping a rather decent diversification and maintaining benchmark-like financial characteristics.

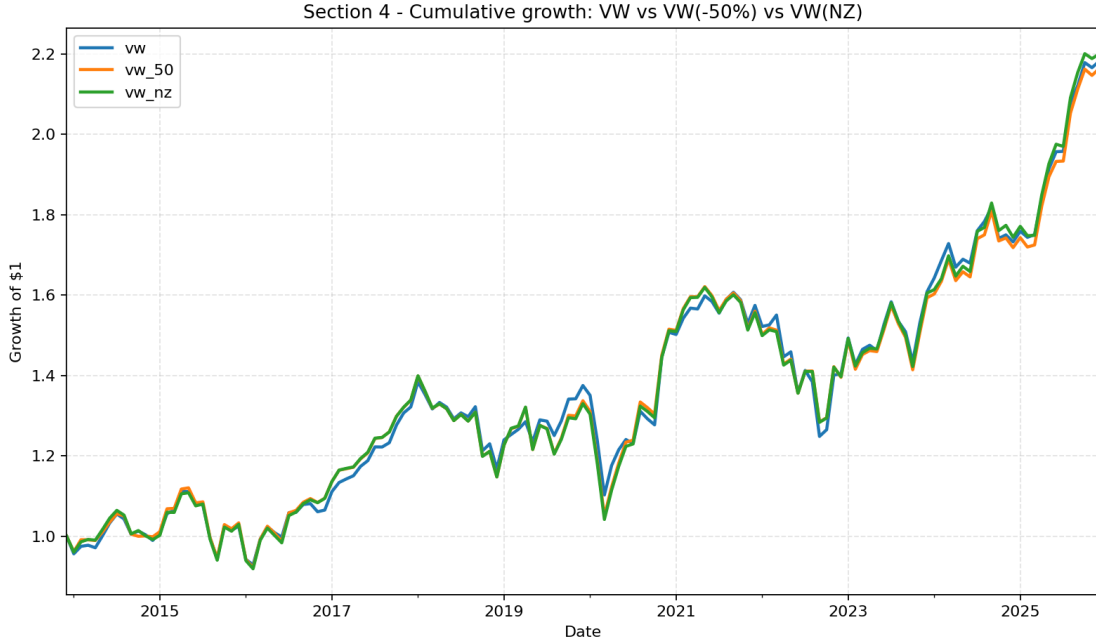


Figure 5: Cumulative performance of the benchmark (VW), the 50% carbon-constrained benchmark-relative portfolio (VW_{50}) and the Net-Zero portfolio (VW_{NZ}). The three paths are visually indistinguishable, indicating that the climate constraint is not strongly priced over this sample.

The economic interpretation is encouraging for a emission-reduction-oriented investor: a credible decarbonisation policy can be implemented in a credible investment strategy. Table 4 shows that the realised footprint of VW_{NZ} remains below or equal to the carbon limit in every selected year.

However, the slack column also shows that the constraint becomes increasingly tight. Apart from 2014, and 2018 where the realised footprint is slightly below the cap, the constraint is exactly binding from 2017 onward. This suggests that the Net-Zero trajectory remains feasible over the sample, but with almost no remaining flexibility by the end of the period.

4.5 Summary tables and economic interpretation

We now consolidate the financial, carbon and tracking-error metrics in four summary tables and discuss what they reveal economically.

Table 1: Financial performance summary

Portfolio	Return (%)	Volatility (%)	Return/Vol	Sharpe	TE vs VW (%)	Cumulative return
<i>VW</i>	7.35	12.89	0.57	0.43	–	1.18
<i>MV</i>	9.14	11.99	0.76	0.61	10.00	1.73
<i>MV_50</i>	8.88	12.75	0.70	0.56	10.60	1.62
<i>VW_50</i>	7.36	13.45	0.54	0.42	2.91	1.16
<i>VW_NZ</i>	7.50	13.49	0.57	0.43	2.86	1.20

Note: The table reports annualised returns, volatilities, the ratio of return to volatility, excess Sharpe ratios (excess return divided by volatility) and ex-ante tracking error versus the value-weighted benchmark. Cumulative return is the total return over the 2014–2025 period, equal to terminal wealth minus one. The returns and volatilities are annualised based on monthly data. Tracking error is reported in percentage points per annum. A dash indicates that the benchmark’s tracking error is zero by definition.

Table 1 reveals a clear ranking in risk-adjusted performance. The unconstrained minimum-variance portfolio (*MV*) delivers the strongest results, with an annualised return of 9.14%, volatility of 11.99%, and an excess Sharpe ratio of 0.61. This outperformance relative to the value-weighted benchmark (*VW*), which returns 7.35% with volatility of 12.89% and an excess Sharpe ratio of 0.43, is consistent with the low-volatility effect documented in the literature.

Adding the 50% carbon constraint to *MV* reduces annualised return from 9.14% to 8.88% and lowers the excess Sharpe ratio from 0.61 to 0.56. The cost is therefore moderate but economically visible, reflecting the loss of flexibility in the optimisation. In contrast, the benchmark-relative strategies remain very close to *VW*: *VW_50* and *VW_NZ* deliver annualised returns of 7.36% and 7.50%, with excess Sharpe ratios of 0.42 and 0.43 respectively. This suggests that carbon constraints can be introduced into a passive benchmark with limited impact on financial performance.

Table 2 summarises average carbon footprints and WACI values. All quantities are expressed in tonnes of CO₂ equivalent per million US dollars (tCO₂e/MUSD).

Table 2: Average carbon metrics across 2014–2025

Portfolio	Carbon footprint (tCO ₂ e/MUSD)	WACI (tCO ₂ e/MUSD revenue)
<i>VW</i>	173	285
<i>MV</i>	441	1 100
<i>MV_50</i>	221	1 160
<i>VW_50</i>	84	355
<i>VW_NZ</i>	82	345

Note: The table reports the average carbon footprint and weighted average carbon intensity (WACI) across all implementation years. The footprint measures the portfolio’s emissions per unit of capital invested, while WACI measures emissions per unit of revenue. Values are rounded to the nearest tonne. The higher WACI of *MV_50* relative to *MV* illustrates that restricting the carbon footprint can concentrate the portfolio into fewer high-intensity firms even though total emissions are reduced.

Table 2 highlights the strong divergence between financial optimisation and climate objectives. The unconstrained minimum-variance portfolio (*MV*) is substantially more carbon-intensive than

the benchmark on both measures: its average carbon footprint reaches 441 tCO₂e/MUSD invested versus 173 for *VW*, while its WACI rises to 1 100 tCO₂e/MUSD revenue compared with 285 for the benchmark. This confirms that pure minimum-variance optimisation tends to allocate heavily toward low-volatility sectors such as utilities, energy and materials, which are often associated with high carbon intensity.

Imposing the 50% carbon constraint significantly reduces the footprint of the minimum-variance strategy, from 441 to 221 tCO₂e/MUSD. However, WACI slightly increases from 1 100 to 1 160 tCO₂e/MUSD revenue. This apparent contradiction reflects the difference between the two climate metrics. The footprint constraint targets emissions per dollar invested, whereas WACI measures emissions relative to company revenues. As a result, the optimiser reduces total financed emissions while concentrating exposures in firms that remain carbon-intensive relative to their revenues.

The benchmark-relative portfolios deliver a more balanced climate profile. *VW_50* reduces the carbon footprint by roughly 52% relative to *VW* (173 to 84 tCO₂e/MUSD), while WACI increases only moderately from 285 to 355 tCO₂e/MUSD revenue. Similarly, *VW_NZ* achieves an average footprint of 82 tCO₂e/MUSD with a WACI of 345 tCO₂e/MUSD revenue. These results suggest that benchmark-relative decarbonisation strategies can substantially reduce financed emissions while preserving a climate profile closer to the benchmark than fully optimised minimum-variance portfolios.

Table 3 reports average ex-ante and ex-post annualised tracking errors relative to the benchmark. The minimum-variance portfolios exhibit high ex-ante tracking error because they minimise total variance rather than active risk. By contrast, the benchmark-relative portfolios achieve very low ex-ante TE by construction, although realised TE is higher because covariances change over time.

Table 3: Average annualised tracking error

Portfolio	Ex-ante TE (%)	Ex-post TE (%)
<i>VW</i>	0.0	–
<i>MV</i>	11.1	10.0
<i>MV_50</i>	10.9	10.6
<i>VW_50</i>	0.1	2.9
<i>VW_NZ</i>	0.1	2.8

Note: Ex-ante tracking error is computed using the estimated covariance matrix at the allocation date, while ex-post tracking error is computed from realised active returns over the implementation period. Values are annualised and expressed in percentage points. The benchmark (*VW*) has zero tracking error by definition.

Table 3 highlights the structural divide between absolute-risk and benchmark-relative optimisations. The minimum-variance portfolios (*MV*, *MV_50*) sit around 11% TE ex-ante and ex-post — high but coherent with their objective, which ignores benchmark proximity. The two numbers are roughly in line, meaning the covariance matrix delivers a fair ex-ante forecast of total active

risk for these portfolios.

The benchmark-relative portfolios are a different story. Ex-ante TE is quasi-null (0.1%): by construction, the optimiser locates a portfolio that is virtually identical to the benchmark in the eyes of Σ_Y . Ex-post TE, however, jumps to 2.9% (*VW_50*) and 2.8% (*VW_NZ*). This 30-fold widening reflects three forces working in the same direction: (i) the covariance matrix is estimated with noise and does not capture future structural breaks; (ii) the carbon constraint tilts the portfolio away from the benchmark in ways that the in-sample optimiser deems risk-free but that realise as active risk; and (iii) the rebalancing-frequency mismatch between the monthly *vw* benchmark and the annually-rebalanced active positions mechanically introduces drift over the implementation year. The takeaway for the investor is that ex-ante TE should be treated as a *lower bound* on realised active risk in any decarbonised benchmark-relative strategy.

Finally, Table 4 summarises selected years of the Net-Zero carbon cap path. The realised carbon footprint of *VW_NZ* always lies below or at the cap, indicating that the optimisation successfully meets the tightening targets.

Table 4: Net-Zero target versus realised carbon footprint

Implementation year	Carbon limit (tCO ₂ e/MUSD)	Realised footprint (tCO ₂ e/MUSD)	Slack (tCO ₂ e/MUSD)
2014	143.04	141.35	1.70
2017	104.28	104.28	0.00
2020	76.02	76.02	0.00
2023	55.42	55.42	0.00
2025	44.89	44.89	0.00

Note: The carbon limit is computed from the benchmark’s 2013 carbon footprint, declining by 10% each year. The realised footprint of the Net-Zero portfolio stays below the limit in all selected years. Slack is the difference between the limit and the realised footprint. Values are expressed in tCO₂e per million US dollars invested.

Table 4 confirms the feasibility of the Net-Zero trajectory over the sample period. The realised carbon footprint of the *VW_NZ* portfolio remains below or equal to the target cap in every selected year. The constraint becomes increasingly tight over time as the carbon limit declines from 143.04 tCO₂e/MUSD in 2014 to 44.89 tCO₂e/MUSD in 2025.

The slack values indicate that the carbon constraint is almost always binding. In 2014, the optimiser achieves a footprint slightly below the cap, leaving a limited slack of 1.70 tCO₂e/MUSD. From 2017 onward, the realised footprint matches the target exactly except for year 2018, implying that the optimisation fully exhausts the available carbon budget. Economically, this suggests that progressively stricter decarbonisation targets remain achievable within the benchmark-relative framework, but only at the cost of an increasingly constrained opportunity set.

These results also indicate that maintaining a mechanically declining Net-Zero pathway may become more challenging beyond the sample period if the investment universe does not continue to decarbonise. Future feasibility would therefore depend on further reductions in firms’ emissions

intensities or on an expansion of the low-carbon investment universe.

5 Discussion

The empirical evidence suggests that climate-aware portfolio construction can be integrated into the asset-management process without necessarily imposing prohibitive financial costs. However, the results also show that the effect of decarbonisation depends strongly on the portfolio objective, the reference benchmark, and the carbon metric used. The main lesson from the analysis is therefore not that climate constraints are always costless, but rather that their cost and interpretation vary substantially across portfolio construction methods.

A first important result is that the unconstrained minimum-variance portfolio (MV) delivers the strongest financial performance. It achieves an annualised return of 9.14%, compared with 7.35% for the value-weighted benchmark (VW), while also displaying lower volatility. This is consistent with the idea that low-volatility stocks can generate attractive risk-adjusted performance. However, this financial improvement comes with a clear climate cost. The average carbon footprint of MV is 441 tCO₂e/MUSD, compared with 173 for VW , and its WACI reaches 1 100 tCO₂e/MUSD revenue, compared with 285 for the benchmark. Pure risk-based optimisation is therefore not climate-aligned by construction: it can improve financial performance while increasing exposure to carbon-intensive firms.

The 50% carbon constraint shows that decarbonisation is feasible, but not neutral. In the minimum-variance case, the footprint falls from 441 to 221 tCO₂e/MUSD, almost exactly achieving the intended reduction. The financial cost remains moderate, with annualised return decreasing from 9.14% to 8.88% and the excess Sharpe ratio falling from 0.61 to 0.56. This suggests that the carbon constraint removes some degrees of freedom from the optimiser, but does not destroy the main financial advantage of the minimum-variance strategy. In contrast, the benchmark-relative portfolios experience a much smaller financial impact. VW_{50} delivers a return of 7.36%, almost identical to VW , while maintaining an ex-post tracking error of about 2.9%. This indicates that carbon reduction is easier to implement when the strategy remains anchored to a diversified benchmark.

A second major finding is that carbon footprint and WACI do not necessarily move in the same direction. For MV_{50} , the carbon footprint falls sharply from 441 to 221 tCO₂e/MUSD, but WACI increases from 1 100 to 1 160 tCO₂e/MUSD revenue. This is not a contradiction. The footprint constraint limits emissions per dollar invested, whereas WACI measures emissions relative to company revenues. As a result, the optimiser can reduce financed emissions while still holding firms that remain carbon-intensive relative to their revenues. This confirms that investors should not rely on a single climate metric: a portfolio may look cleaner on a financed-emissions basis while still carrying high revenue-based carbon intensity.

The benchmark-relative climate strategies provide the most balanced outcome. VW_{50} reduces the average carbon footprint from 173 to 84 tCO₂e/MUSD, while VW_{NZ} reaches 82 tCO₂e/MUSD.

At the same time, their cumulative performance remains close to the benchmark and their tracking errors stay below 3%. These results suggest that substantial financed-emissions reductions can be achieved without materially changing the financial behaviour of a passive benchmark portfolio. Nevertheless, WACI rises to 355 for *VW_50* and 345 for *VW_NZ*, compared with 285 for *VW*. This reinforces the idea that benchmark-relative decarbonisation is effective on footprint reduction, but does not automatically improve every dimension of climate exposure.

The Net-Zero strategy is also feasible over the sample period, but the feasibility margin becomes very tight. The realised footprint of *VW_NZ* remains below or equal to the carbon cap in every selected year. However, the slack is positive only in 2014 and equals zero from 2017 onward. This means that the optimiser fully uses the available carbon budget in most implementation years. Economically, this is encouraging because the Net-Zero path can be implemented within the Pacific equity universe during 2014–2025. At the same time, it also suggests that continuing the same mechanical decarbonisation trajectory beyond the sample could become more difficult unless the investment universe itself continues to decarbonise.

Overall, the results support a nuanced conclusion. Climate-aware portfolio construction is feasible and can deliver large reductions in financed emissions with limited financial disruption, especially when implemented relative to a benchmark. However, the results should not be interpreted as evidence that decarbonisation mechanically improves performance. They are conditional on the Pacific universe, the 2014–2025 sample period, annual rebalancing, the covariance estimation method, the risk-free rate alignment, and the absence of transaction costs. Future robustness checks should therefore examine alternative carbon thresholds, joint footprint and WACI constraints, sector-neutral constraints, transaction costs, turnover limits and alternative covariance estimators. These extensions would help determine whether the results are stable or partly driven by sample-specific sector composition and market conditions.

6 Robustness Checks and Possible Improvements

Although the results are informative, several robustness checks would be needed before drawing strong conclusions about the general performance of climate-aware portfolio construction. The current analysis shows that decarbonisation is feasible, especially for benchmark-relative portfolios, but it also reveals methodological issues related to carbon metrics, covariance estimation, concentration and implementation costs. The following extensions would improve the reliability and practical relevance of the analysis.

- **Test alternative carbon-reduction thresholds.** The current analysis focuses mainly on a 50% carbon-footprint reduction and a 10% annual Net-Zero decarbonisation path. A natural extension would be to test additional thresholds, such as 25%, 50% and 75% footprint reductions, as well as alternative Net-Zero decay rates such as 5%, 10% and 15% per year. This would help identify the point at which the carbon constraint becomes too restrictive. This is particularly relevant because Table 4 shows that the Net-Zero constraint is binding

or nearly binding from 2017 onward, indicating that stricter targets could quickly reduce the feasible set.

- **Add a joint WACI constraint.** One of the main findings of the report is that carbon footprint and WACI do not necessarily move together. In particular, *MV 50* reduces its carbon footprint from 441 to 221 tCO₂e/MUSD, while its WACI increases from 1,100 to 1,160 tCO₂e/MUSD revenue. This shows that constraining financed emissions alone may still leave the portfolio exposed to firms that are carbon-intensive relative to revenues. A useful extension would therefore be to impose an additional WACI constraint, for example requiring WACI to remain below the benchmark level or to decrease by a fixed percentage.
- **Use sector-neutral carbon constraints.** The carbon exposure of the minimum-variance portfolios may partly reflect sector tilts toward low-volatility but carbon-intensive industries. A sector-neutral version of the carbon constraint would separate two effects: reductions obtained by moving away from carbon-heavy sectors, and reductions obtained by selecting cleaner firms within each sector. This would make the economic interpretation more precise and reduce the risk that the results are mainly driven by sector allocation.
- **Improve covariance estimation.** The optimisation relies on a historical rolling covariance matrix, which can be noisy and may underestimate future active risk. This is visible in Table 3, where the benchmark-relative portfolios have very low ex-ante tracking error but realised tracking error around 2.8–2.9%. Future work could compare the current estimator with Ledoit–Wolf shrinkage, EWMA covariance matrices, or factor-model covariance estimators. More stable covariance estimates could improve the reliability of ex-ante risk and tracking-error forecasts.
- **Include transaction costs and turnover constraints.** The current backtest assumes frictionless rebalancing. In practice, reducing carbon exposure requires selling some carbon-intensive stocks and reallocating toward cleaner firms, which creates transaction costs, bid–ask spreads and potential market-impact costs. Adding a turnover penalty or a maximum turnover constraint would make the strategies more realistic and would show whether the financial performance remains attractive after implementation costs.
- **Measure concentration and diversification.** Carbon constraints can reduce the feasible investment universe and may increase concentration. The report could therefore be strengthened by adding diversification indicators such as the effective number of holdings,

$$N_{\text{eff}} = \frac{1}{\sum_i w_i^2},$$

and the Herfindahl–Hirschman index,

$$HHI = \sum_i w_i^2.$$

These measures would show whether carbon reductions are achieved through broad reallocations or through concentration in a limited number of low-carbon firms.

- **Run sub-period and factor-attribution tests.** The sample includes different market regimes, including the pre-COVID period, the COVID shock and the post-2020 recovery. Repeating the analysis over sub-periods would show whether the results are stable across market environments. In addition, a multi-factor regression using factors such as market, size, value, momentum, quality and low volatility would help identify whether the strong performance of *MV* and *MV 50* is mainly due to low-volatility exposure rather than to the carbon-aware construction itself.
- **Check sensitivity to carbon-data assumptions.** Carbon metrics depend on data availability, revenue definitions, market values and emissions reporting quality. Future robustness checks should test the impact of alternative missing-data treatments, winsorisation of extreme carbon intensities, and the inclusion of Scope 3 emissions when available. This would strengthen the credibility of the climate results, especially because WACI and carbon footprint react differently to data assumptions.

Overall, the most important improvements would be to test alternative carbon targets, add a joint WACI constraint, improve covariance estimation, and include transaction costs and concentration measures. These extensions directly address the main issues identified in the empirical results: the divergence between carbon footprint and WACI, the increasingly tight Net-Zero constraint, the gap between ex-ante and realised tracking error, and the practical implementation cost of decarbonisation.

7 Limitations

The results should be interpreted with caution, since they depend on the quality of the data, the portfolio constraints and the assumptions used in the backtest. The purpose of this section is not to undermine the results, but to clarify the conditions under which they hold.

A first limitation concerns carbon data. Emissions and revenues are reported annually, while returns are measured monthly. The portfolio is therefore optimised using the most recent annual carbon information available at the allocation date, but performance is then evaluated month by month over the following year. This frequency mismatch is unavoidable in this type of exercise, but it means that changes in firms' emissions during the year are not immediately reflected in the allocation. In addition, firms with missing emissions, missing revenues, unreliable market capitalisation or insufficient return history are excluded from the investable universe. This cleaning step is necessary, but it may create a selection bias if firms with poor disclosure are systematically different from firms with complete data.

A second limitation comes from covariance estimation. The optimisation relies on a rolling historical covariance matrix estimated from monthly returns, with a small ridge adjustment for

numerical stability. This estimator is transparent and easy to implement, but it remains noisy. The issue is visible in Table 3: the benchmark-relative portfolios have an ex-ante tracking error close to 0.1%, but their realised tracking error reaches about 2.9% for *VW_50* and 2.8% for *VW_NZ*. This gap suggests that the covariance matrix identifies portfolios that look very close to the benchmark in-sample, but that this proximity is weaker out-of-sample.

A third limitation is the absence of stronger diversification constraints. The portfolios are long-only and fully invested, but the optimisation does not impose maximum stock weights, sector neutrality, turnover limits or ownership constraints. This matters especially for the minimum-variance portfolios. The unconstrained *MV* portfolio already has an ex-post tracking error of about 10% relative to the benchmark, and the carbon-constrained version *MV_50* remains similarly far from the benchmark, with an ex-post tracking error of 10.6%. Therefore, even though *MV_50* reduces the carbon footprint from 441 to 221 tCO₂e/MUSD, the strategy remains an active and potentially concentrated portfolio.

The model also abstracts from liquidity and implementation costs. The optimisation is expressed only in weights, without checking whether the corresponding positions would be realistic for small or illiquid firms. In practice, large allocations to small-cap firms could generate market impact or ownership issues. Moreover, the reported returns are gross of transaction costs, bid-ask spreads, taxes and rebalancing costs. Since carbon-constrained portfolios need to adjust their weights in order to remain below their carbon limits, their net performance could be lower once implementation costs are included.

The Net-Zero result is also conditional on the sample period. Table 4 shows that the *VW_NZ* portfolio satisfies the carbon cap in every selected year. However, the constraint becomes increasingly tight: the carbon limit falls from 143.04 tCO₂e/MUSD in 2014 to 44.89 tCO₂e/MUSD in 2025, and the slack is almost zero from 2017 onward. This suggests that the Net-Zero path is feasible in the historical sample, but that its continuation beyond 2025 would depend on whether the investment universe itself continues to decarbonise.

Finally, portfolio decarbonisation should not be confused with real-economy decarbonisation. A lower financed carbon footprint reflects a reallocation of capital toward lower-footprint firms, but it does not directly imply that firms reduce their actual emissions. The results therefore provide evidence on portfolio-level carbon reduction, not on causal reductions in real-world emissions. A complete climate investment policy would need to combine portfolio optimisation with other tools, such as engagement, voting policies, transition-plan analysis and sector-level climate assessment.

8 Conclusion

This report studies whether carbon-aware portfolio construction can reduce the climate exposure of a Pacific equity portfolio without materially weakening its financial performance. The analysis compares five strategies over the 2014–2025 period: the value-weighted benchmark (*VW*), the unconstrained minimum-variance portfolio (*MV*), the carbon-constrained minimum-variance

portfolio (*MV_50*), the benchmark-relative 50% carbon-constrained portfolio (*VW_50*), and the benchmark-relative Net-Zero portfolio (*VW_NZ*).

The first result is that the unconstrained minimum-variance strategy performs well financially, but is not climate-aligned. As shown in Table 1, *MV* delivers an annualised return of 9.14%, compared with 7.35% for *VW*, while also having lower volatility, 11.99% versus 12.89%. Its excess Sharpe ratio is therefore higher, at 0.61 compared with 0.43 for the benchmark. However, this improvement in risk-adjusted performance comes with a much higher carbon exposure. Table 2 shows that the average carbon footprint of *MV* is 441 tCO₂e/MUSD, compared with 173 tCO₂e/MUSD for *VW*. This confirms that a portfolio optimised only on variance may allocate heavily to low-volatility but carbon-intensive firms.

The second result is that imposing a carbon constraint on the minimum-variance portfolio achieves the intended climate objective with a moderate financial cost. The *MV_50* portfolio reduces the average footprint from 441 to 221 tCO₂e/MUSD, almost exactly a 50% reduction. At the same time, its annualised return declines only from 9.14% to 8.88%, and its excess Sharpe ratio falls from 0.61 to 0.56. The cost of decarbonisation is therefore visible, but not prohibitive. The main trade-off is rather that *MV_50* remains far from the benchmark, with an ex-post tracking error of 10.6%, which points to higher active risk and potentially stronger concentration.

The benchmark-relative strategies provide a different picture. *VW_50* reduces the average carbon footprint from 173 to 84 tCO₂e/MUSD, while keeping annualised return almost unchanged at 7.36%, compared with 7.35% for *VW*. Its excess Sharpe ratio is also close to the benchmark, 0.42 versus 0.43, and its realised tracking error remains limited at about 2.9%. This suggests that a benchmark-oriented investor can achieve a large reduction in financed emissions without moving too far away from the benchmark.

The Net-Zero portfolio leads to a similar conclusion. The *VW_NZ* strategy follows a carbon cap declining by 10% per year and remains below or equal to the target in every selected year. Financially, it remains close to the benchmark, with an annualised return of 7.50%, volatility of 13.49%, and an excess Sharpe ratio of 0.43. Its realised tracking error is 2.8%, close to that of *VW_50*. These results suggest that a gradual Net-Zero pathway can be implemented within a benchmark-relative framework without generating a large performance penalty over the sample period.

Overall, the results show that the cost of decarbonisation depends on the investor’s objective. For an absolute-risk optimiser, the carbon constraint is more restrictive because it directly changes the minimum-variance solution. For a benchmark-relative investor, the cost is smaller because the optimisation is designed to preserve proximity to the benchmark while reducing carbon exposure. This explains why *MV_50* remains financially attractive but carries high active risk, whereas *VW_50* and *VW_NZ* achieve large carbon reductions with much lower tracking error.

The analysis also highlights the need to distinguish carbon footprint from WACI. The carbon footprint constraint successfully reduces emissions per dollar invested, but this does not automatically imply a proportional reduction in emissions intensity relative to revenues. For example,

MV_50 halves the footprint of *MV*, while its WACI remains slightly higher. This shows that investors should not rely on a single climate metric, but should define clearly whether the objective is to reduce financed emissions, revenue-based carbon intensity, or both.

In conclusion, the project shows that climate objectives can be integrated into portfolio construction in a disciplined quantitative framework. The strongest evidence comes from the benchmark-relative strategies: *VW_50* and *VW_NZ* deliver substantial carbon footprint reductions while preserving returns, Sharpe ratios and tracking error close to the benchmark. However, these findings remain conditional on the Pacific universe, the 2014–2025 period, the covariance estimator and the absence of transaction costs and liquidity constraints. Carbon-aware investing is therefore feasible, but its implementation requires careful control of diversification, active risk, trading costs and the distinction between portfolio decarbonisation and real-world emissions reduction.

9 Appendix

A Additional Portfolio Weights

Table 5 summarises the size of each portfolio and lists the top holdings ranked by average weight over the full sample (2013–2025). The average number of holdings is computed across annual rebalancing dates. For brevity the table reports the four largest positions for each strategy; the full holdings data are available in the accompanying CSV file.

Table 5: Summary of portfolio holdings and top positions

Portfolio	Avg. holdings	Top position	2nd position
<i>MV</i>	29.2	ITOCHU TECHNO-SOLUTIONS (16.3%)	CLP HOLDINGS (14.6%)
<i>MV_50</i>	30.0	ITOCHU TECHNO-SOLUTIONS (18.0%)	NTT (15.9%)
<i>VW_50</i>	291.5	DBS GROUP HOLDINGS (3.7%)	HANG SENG BANK (2.7%)
<i>VW_drift</i>	365.2	TOYOTA MOTOR (4.9%)	COMMONWEALTH BK. OF AUS. (2.7%)
<i>VW_NZ</i>	288.8	DBS GROUP HOLDINGS (3.8%)	HANG SENG BANK (2.8%)

Note: The table reports the average number of holdings (across rebalancing dates) and the four largest positions for each portfolio value. The full time series of holdings and weights is provided in the accompanying CSV file.

B Top WACI Contributors

To illustrate why weighted average carbon intensity (WACI) may increase even when the financed carbon footprint falls, Table 6 lists the ten companies with the highest average contribution to the benchmark’s WACI over the sample. Utilities and energy firms dominate the list, with the top contributor, Power Assets Holdings, responsible for more than 140 tCO₂e per million dollars of revenue on average.

Table 6: Top contributors to WACI in the benchmark (2013–2025)

Company	Average WACI contribution (tCO ₂ e/MUSD revenue)
Power Assets Holdings	140.22
CKI Holdings	32.21
China Resources Power	18.29
CLP Holdings	17.57
Woodside Energy Group	8.88
ELEC. Power Dev.	8.44
Swire Pacific 'A'	6.86
Chin. Pwr. Intdvt.	6.20
BHP Group	5.76
Nippon Steel	5.74

Note: The table reports the mean WACI contribution of each company to the value-weighted benchmark across 2013–2025. Contributions are measured in tCO₂e per million dollars of revenue.

C Tracking Error Illustration

Figure 6 visualises the ex-ante (modelled) and ex-post (realised) tracking error of the Net-Zero strategy relative to the benchmark. The realised tracking error is materially higher than the ex-ante estimate, supporting the recommendation to use more robust covariance estimators. The figure is placed in the appendix to avoid breaking the flow of the main text.

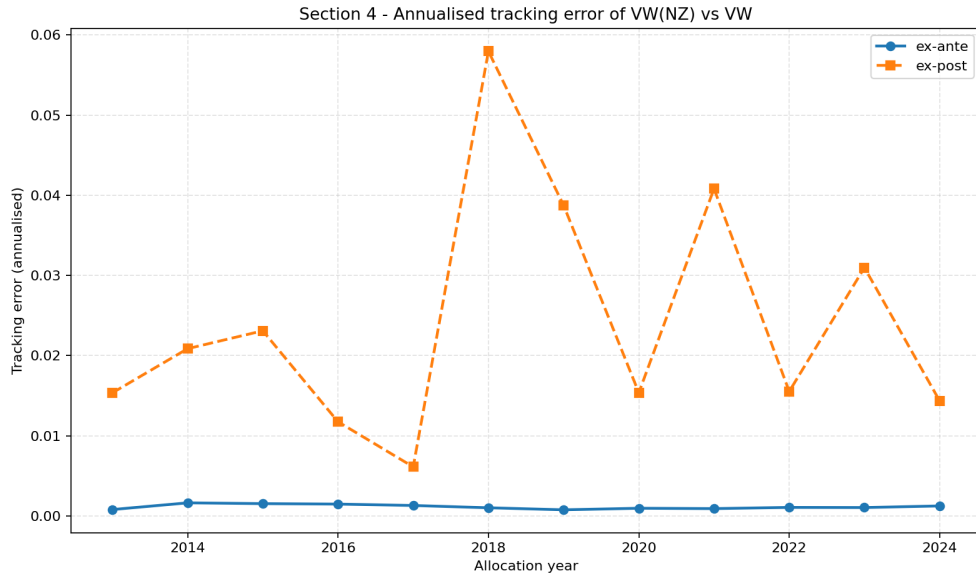


Figure 6: Ex-ante vs ex-post tracking error of the Net-Zero portfolio relative to the benchmark

D WACI Dynamics Illustration

Figure 7 compares the weighted average carbon intensity of the benchmark, the 50% carbon-constrained strategy and the Net-Zero strategy. Even though both constrained portfolios achieve large reductions in financed emissions, WACI does not decline mechanically, especially in the minimum-variance setting. This underscores the need to monitor both footprint and intensity when designing climate-aware portfolios.

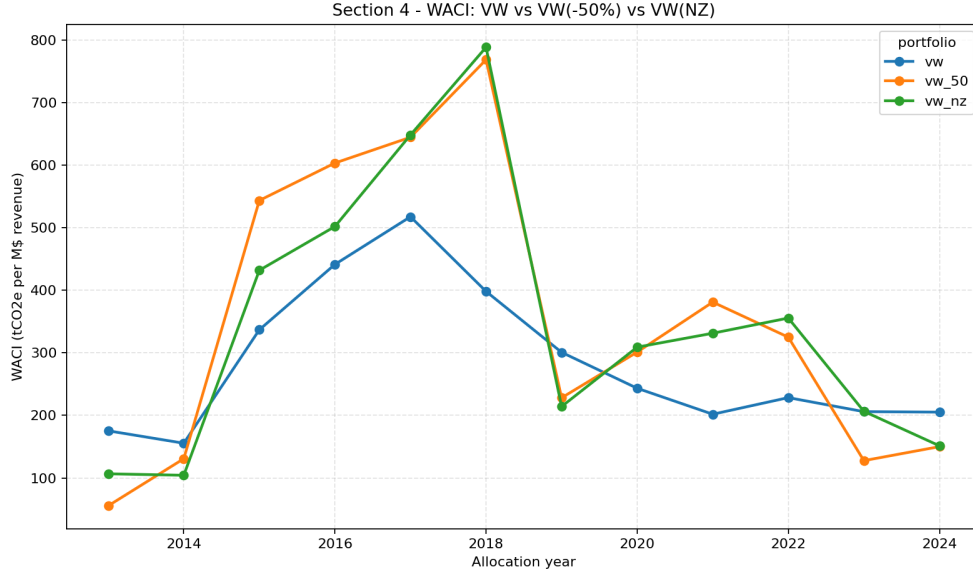


Figure 7: WACI of the benchmark, 50% carbon-constrained and Net-Zero portfolios

E Related Literature

This project is closely related to three strands of literature. First, the portfolio construction follows the mean–variance framework introduced by Markowitz (1952), where investors allocate across assets by trading off expected return and risk. The use of minimum-variance portfolios also connects to the literature on the low-volatility anomaly, which documents that low-risk stocks can deliver strong risk-adjusted performance. This is consistent with the empirical finding in this report that the unconstrained minimum-variance portfolio outperforms the value-weighted benchmark financially, although it is not climate-aligned.

Second, the project relates to the literature on constrained portfolio optimisation. Constraints such as long-only weights, benchmark-relative tracking error, and carbon limits can materially affect portfolio composition and realised performance. This is particularly relevant in our results, where the carbon constraint reduces the footprint of the minimum-variance portfolio but also limits the optimiser’s flexibility and creates a divergence between carbon footprint and WACI.

Third, the report contributes to the sustainable investing and climate-aware portfolio literature. Prior studies on decarbonised and low-carbon portfolios show that investors can reduce financed

emissions while maintaining a portfolio close to a benchmark. Our results are consistent with this view for the benchmark-relative portfolios, since *VW_50* and *VW_NZ* strongly reduce carbon footprint while keeping tracking error around 2.9%. However, the increase in WACI for several constrained portfolios confirms that climate alignment depends strongly on the metric used. This supports the idea that investors should monitor both financed emissions and revenue-based carbon intensity when assessing the climate profile of a portfolio.

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Use of Large Language Models

Large Language Models were used as supporting tools during this project. Claude (Anthropic) was used as a coding assistant to review the notebook structure, debug implementation issues, merge different notebook versions, and improve code readability. ChatGPT (OpenAI) was also used to improve the clarity of the written report, check numerical consistency between tables and interpretations, and refine the discussion of the results. The LLMs were not used to generate portfolio results independently, and all numerical outputs, methodological choices, economic interpretations, and final academic responsibility remain with the group.